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Communicating uncertainty to policy makers in maps: A systematic scoping review of meaningful to misleading current practice

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Abstract

Policymakers frequently use maps as decision-support tools to support the effective allocation of resources and enhance community resilience against climate-related health threats. Yet, the high legibility of maps can be accompanied by the risk of inappropriate or misleading abstraction of uncertainty. This systematic scoping review explored how uncertainty was communicated in geographical visualisations present in peer-reviewed publications on climate-related infectious diseases in the Asia-Pacific region. A PRISMA-compliant systematic scoping review was carried out in PubMed and Web of Science databases in April 2025. Manuscripts describing primary research on climate-sensitive human diseases in the Asia-Pacific region, and with one or more map, were included. The extent to which uncertainty was explicitly represented, visually within figures or textually in figure captions, was qualitatively assessed and narratively synthesized. Sixty-six manuscripts were identified from an original 2,826 search results, most ($n = 52$) of which indicated policymakers, public health authorities, or health practitioners were intended target audiences. Eleven studies reported uncertainty in maps textually, and only three studies explicitly integrated explanatory visual representations of uncertainty into maps. Our findings suggest that most of the current policy-facing literature regarding climate-related infectious disease in the Asia-Pacific region does not employ appropriate visual methods to convey uncertainty in maps. There is an urgent need to improve current practice in the visual communication of uncertainty to support informed, considered policy and planning relating to climate-related infectious disease.

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1. Introduction

There is an urgent need to increase the global population's preparedness for the ongoing and projected impacts of climate change [1]. While much research on the impacts of climate on infectious disease focuses on North America and Europe, there is a growing body of evidence regarding the Asia Pacific region [2]. The Asia-Pacific

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carries a disproportionate burden of climate-sensitive infectious diseases, with over 70% of its population estimated to be at risk [3]. It is well recognised that the impacts of climate change require an informed, iterative and coordinated policy response [4]. Historically, climate-related policy across the Asia Pacific region has been characterised by fragmentation and limited evaluation [5,6]. This is changing through efforts such as the Association of South-East Asian Nations coordination of health system adaptation in response to climate change-related infectious disease [7]. Maps are frequently used as decision-support tools to support climate-related health policy, yet much of our understanding of current and best practice is derived from an American or European policy context (e.g., Terrado et al. [8]). It remains unclear if current research regarding maps for integrating evidence into policy undertaken elsewhere applies in the Asia Pacific context.

Maps combine visual clarity, spatial context, and statistical depth to enhance understanding of the underlying information [9]. Various types of maps are used in population health geographics, including choropleth maps, heat maps, thematic maps, topographic maps, dot density maps, and cartograms [10]. Over time, mapping has evolved from static representations to sophisticated tools capable of real-time data visualization, predictive modeling, and multi-layered analysis. These diverse mapping techniques enhance research, policy, and public understanding. Policymakers have increasingly utilized maps to identify spatial patterns, track disease outbreaks, and allocate resources effectively [11]. Maps can be used to anticipate disease outbreaks linked to climatic events, such as floods or extreme temperature changes, assist in the identification of vulnerable populations, track disease transmission patterns, and assess environmental risk factors like rainfall, temperature fluctuations, or land-use changes. e.g., as in Delmelle et al. [11].

Uncertainty is a complex, socially-constructed concept [12]. Applying the taxonomy of unknowns from Smithson 1989, the current work focuses on uncertainty as a consequence of error, that is, incomplete or distorted knowledge. Incompleteness can manifest as vagueness or ambiguity, which statistical analyses commonly aim to clarify [13]. Distortion can arise from confusion, such as mistaking one thing for another (e.g., false negative or positive test results), or inaccuracy (e.g., lack of precise records of the location relevant to a piece of data). Policymakers operate under time, financial, and political pressures that require efficient and correct grasp of uncertainties both in the information they are provided, and in projected consequences of any given decision [14]. Understanding the degree of uncertainty in any given piece of information can help policymakers focus more on reliable evidence, separate meaning from noise, and direct resources to reduce uncertainty through additional data collection. Importantly, failure to appropriately convey and understand uncertainty can lead to “incredible certitude”, where past evidence and/or future predictions are taken as more accurate and precise than is correct, or even possible [15].

Most audiences, from the inexperienced to domain experts, find it difficult to intuitively understand uncertainty. Visualisation can help address this [16]. The goal of visualising uncertainty is to represent information as accurately as possible, which includes representing error intrinsic to underlying data and/or models [17]. Although

the importance of visualising uncertainty to support decision making is highly valuable, the variety of potential underlying uncertainties, and the methods available to visualise, them are ongoing challenges to best practice [18]. Maps are subject to uncertainties arising from considerations of data accuracy and the visualization techniques chosen. Positional uncertainty arises from inaccuracies in geographic coordinates (location), while attribute uncertainty stems from incomplete, misclassified, or outdated data (data quality) [11,19]. Scale and aggregation uncertainty, known as the Modifiable Areal Unit Problem, distorts spatial patterns when data is aggregated at different spatial scales, masking local patterns [20]. Temporal uncertainty occurs when maps fail to capture real-time changes, leading to outdated representations of dynamic phenomena such as climate change or disease outbreak progression [11].

Uncertainty can be incorporated into geographical communication in a number of ways. This includes within the visualisation itself, such as confidence intervals, transparency overlays, blurred borders, dashed rather than solid lines, and probabilistic forecasting. These allow users to understand the inherent limitations of spatial data rather than assuming absolute precision [20,21]. The depiction of uncertainty can also include description and disclosure of uncertainties in the figure caption, or references in the figure caption to other areas of text which describe uncertainties, such as tables reporting estimates alongside standard error.

This systematic scoping review aims to explore how uncertainties are communicated in geospatial maps used in climate-related infectious disease research across the Asia-Pacific region. It aims to examine how maps present or fail to present uncertainties related to data sources, modelling assumptions, and temporal or spatial precision, and how these practices influence their utility for policy and decision-making. By assessing the extent to which uncertainties are explicitly represented in maps, visually or textually, and whether current mapping practices support informed, cautious, and evidence-based public health responses, we hope to inform future efforts to improve the clarity, reliability, and ethical use of geographical and geospatial visualisations in climate-health policymaking.

2. Methods

2.1. Study design

A systematic scoping review is a research method that involves a structured and comprehensive synthesis of existing literature on a specific topic, using predefined criteria and transparent procedures. Our approach follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) protocol developed prior to commencing data collection and analysis, guided the study's design and methods (S2 File). Briefly, this may be considered a hybrid approach with characteristics of a systematic review (predefined search string, inclusion and exclusion criterion; formal screening process), and a scoping review (search strategy based on refinement of initial exploratory search; focus on current practice in uncertainty visualisation rather than effectiveness of a given method of uncertainty visualisation; qualitative approach to outcome data extraction). The aim was to examine the current literature on communicating uncertainties in geospatial maps related to climate-sensitive infectious diseases and their implications for policymakers.

2.2. Data sources and search strategy

A three-step search strategy was employed to ensure a comprehensive and holistic review of the literature. First, we conducted an initial search in the PubMed database to identify a preliminary set of articles representing relevant literature. Review of these articles and their reference lists were used to establish the final search strategy detailed in Table 1.

Next, we systematically searched peer-reviewed articles in two databases: PubMed and Web of Science, selected for their extensive coverage of biomedical and environmental health literature pertinent to our topic.

The search strings outlined in Table 1 were applied across both databases, in "all fields" (specific syntax was *TS*= for Web of Science, and *[tiab]* for PubMed). Using filters, we limited our search to articles published in English (specific

Table 1. Search strategy.

Key Concept	Search Terms/Keywords
Geographic region	Australia OR "New Zealand" OR Fiji OR "Papua New Guinea" OR "Solomon Islands" OR Vanuatu OR Samoa OR Tonga OR Kiribati OR Micronesia OR "Marshall Islands" OR Nauru OR Tuvalu OR "Cook Islands" OR Niue OR Polynesia OR Melanesia OR "Asia-Pacific" OR Australasia OR Oceania OR "Pacific region" OR "New Caledonia" OR "Lord Howe Island"
	AND
Environmental factors	"climate sensitive" OR temp* OR "climate change" OR environment* OR rain* OR precip*
	AND
Health outcomes	diseas* OR infect*
	AND
Spatial methods	GIS OR Geospatial OR spat* OR "geographic information system*"

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syntax was *LA=(English)* in Web of Science, and *English[lang]* for PubMed). There were no restrictions on publication date. The search included all papers available up to April 2025.

Finally, we conducted a hand search of the reference lists of included papers to identify additional literature that might have been missed during the database searches.

Relevance to the Asia-Pacific region was inferred by explicit mention of terms used for countries, subregions, and broad regional labels commonly used in the relevant infectious diseases literature pertaining to Pacific Island countries and territories, Australia, and New Zealand (Table 1, line 1). South Asia, Southeast Asia, and South Asia were not included in the search terms to limit the number of records to something feasible for a scoping study.

2.2.1. Inclusion and exclusion criteria. The inclusion and exclusion criteria were defined prior to screening. Studies were included if they met the following criteria: (1) peer-reviewed manuscripts written in English, (2) focused in whole or in part on climate-sensitive infectious diseases and their relationship with geospatial mapping and uncertainty communication, (3) conducted within the Asia-Pacific region, (4) explored communication or interpretation of uncertainties in geospatial data or maps relevant to public health decision-making, and (5) involved human health outcomes.

Studies were excluded if they: (1) focused exclusively on non-human subjects such as vectors or animals without direct implications for human health, (2) were review articles, editorials, commentaries, letters to the editor, conference abstracts, or other non-primary research, (3) were conducted outside the Asia-Pacific region, or (4) were published in languages other than English.

2.2.2. Study selection process. The screening process is summarized in Fig 1. First, results from the two databases were combined, and duplicate records were removed. Titles of all remaining records were screened based on the inclusion and exclusion criteria, with any disagreements resolved through consensus. The same process was applied to the screening of abstracts. Following this, full-text screening was conducted based on the inclusion of GIS maps in the research study. Additionally, to identify relevant studies not captured in the database searches, the reference lists of the studies included for data extraction were also screened.

2.2.3. Data extraction and analysis. Data from all included studies, were extracted by one reviewer (AJ) with 15% blinded double screening verification from EIW, and randomised checks by AL, with any disagreements resolved through consensus. Ambiguous cases (e.g., where low image quality of a given figure may obscure whether a visual feature was intended to be blurred or sharp; or where presence of absence of a given form of ambiguity was unclear) were flagged by

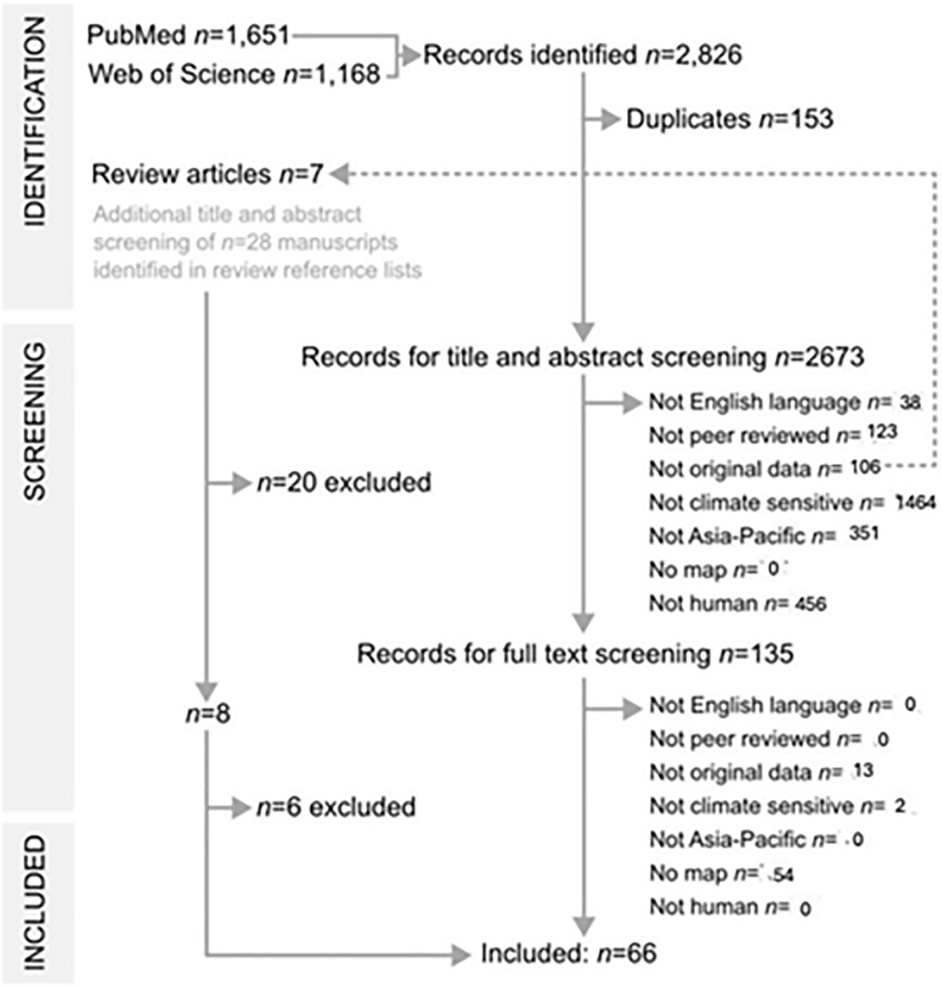


Fig 1. PRISMA flow diagram of manuscript inclusions/exclusions.

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the extractor and resolved through discussion with EIW and AL. As this was a qualitative scoping review, a fully featured data extraction template was not applied; rather, we applied a series of a-priori data features (Table 2) and inductively identified recurring types of uncertainty (Table 3) to guide extraction of salient characteristics of included figures.

A critical qualitative analysis was then conducted on all full-text studies containing maps, based on the above criteria, to evaluate the communication of uncertainties in maps and implications for policymaking.

3. Results

3.1. Overview of included studies

There were 2,826 results from searches. After removing duplicates, applying the inclusion and exclusion criteria, 135 studies from the original search underwent full text screening. The original search yielded seven reviews, from which an additional 28 papers of possible relevance not included in the search results were identified. Following full text screening, 66 papers were included for data extraction. (S1 File).

Table 2. Information extraction.

Variable	Rationale and criterion
Target audience and policy relevance	Identification of who the study was primarily aimed at, such as policymakers or public health officials, indicated by direct quotes where available or inferred from context.
Data source	Sources of data (e.g., government or international repositories), and/or where data was collected, as reported by authors. Data source uncertainty was extracted for each data source. Building on the logic that smaller, more ad-hoc data collection efforts are likely to be more specific but less comprehensive (and thus generalisable), and larger secondary datasets are likely to be more comprehensive and well-documented, this was classified as "high" if the study involved primary data collection, "moderate" if it included a mix of primary and secondary data, and "low" if only secondary data sources were used.
Temporal considerations	This was extracted for each data source. It included when data collection was undertaken, and whether data was cross-sectional or longitudinal. The purpose was to identify uncertainty arising from temporal limitations relating to the timing of data collection or reporting. This may include (1) narrow windows of data collection, or cross-sectional data collection, potentially leading to uncertainty regarding change over time such as seasonality, (2) sparse time points of data collection potentially leading to uncertainty about unmeasured dynamics between sampling occasions; and (3) notable delays between when data was collected and when the paper was published, leaving uncertainty about whether findings relate to the current state of the world.
Visual depiction of uncertainty	This was extracted for each figure meeting inclusion criterion. Derived from qualitative judgement following review of the literature regarding depiction of uncertainty and best practice in figures involving maps, this included any uncertainties observed in the map figures, such as statistical uncertainties displayed in text in figures, or in figure captions. See results for types of uncertainties extracted. Additionally, correspondence between uncertainty disclosed in text and uncertainty depicted in the figure was extracted. This was achieved by review of manuscript methods, and any subsequent text regarding each figure included for visual data extraction (e.g., references to figure X in results or other sections). References to uncertainty in text were noted, and then the figure was scrutinised for presence or absence of disclosure of that uncertainty (e.g., if spatial resolution of data collection instrument is referred to in text, but the figure shows higher resolution without appropriate disclosure of interpolation in the figure or caption).

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All included studies focused on the Asia-Pacific region, consistent with the geographic scope of this systematic scoping review. The time period covered by these studies ranged from 1986 to 2025, with 96.6% published within the last 20 years, reflecting a growing research interest in this field in recent decades.

Among the included studies, the most commonly investigated climate-related infectious diseases were malaria, dengue fever, scrub typhus, hand, foot and mouth disease, and chikungunya virus. Malaria was particularly well-represented with multiple studies focusing on its spatial and temporal distribution across Papua New Guinea, Indonesia, and other Asia-Pacific regions. Dengue fever and other arboviral diseases, such as Ross River virus and Barmah Forest virus, were frequently examined in studies from Australia and Southeast Asia.

Other infectious diseases studied included leptospirosis, influenza A virus, soil-transmitted helminths, and scrub typhus, highlighting the diverse range of climate-sensitive infectious diseases addressed using geospatial mapping in the Asia-Pacific context. This diversity underscores the broad relevance of GIS-based uncertainty communication across multiple public health challenges.

Table 3. Identified recurring types of uncertainty.

			Counts of acknowledged or reported uncertainty (per manuscript)			
Visual uncertainty type	**Definition for coding**	**Examples of failures to represent this form of uncertainty noted in coding**	**Not present**	**Present in text only**	**Present in caption only**	**Present in visuals**
Spatial	Appropriate representation of the spatial extent of data or modelling predictions. Relates to inaccuracy within distortion per Smithson 1989. Focus is on spatial representations rather than aesthetics.	• Inaccuracies in geographical coordinates (positional uncertainty) • Data misclassification (attribute uncertainty) • Aggregation effects, commonly arising from the Modifiable Area Unit Problem • Interpolation assumptions that are not disclosed • Oversimplification of forms (e.g., irregular shapes smoothed to circle plots)	N/A	17	0	3
Statistical	Representation of error intrinsic to data or underlying models. Relates to disclosure of vagueness and ambiguity per Smithson 1989.	• Lack of confidence/credible/prediction intervals • Lack of error bars or uncertainty bands • Lack of disclosure of sample size that may vary across spatial units (e.g., districts, postcodes) • Model-based smoothing that may obscure real features of the data (e.g., shading based on model outputs following outlier removal, without disclosure of outlier removal)	N/A	11	9	2
Visual clarity	Representation of information clearly to the viewer. Relates to confusion within distortion per Smithson 1989, and the principles outlines in Tufte's "The Visual Display of Quantitative Information" [29]. Focus is on aesthetics rather than spatial representations.	• Colours scales that imply discrete nature of a continuous variable, or vice versa • Depiction of hard boundaries where fuzzy or gradient edges would better reflect the underlying data or model • Low-resolution image that obscures detail important to understanding what is being depicted • Use of visual motifs that do not follow convention (e.g., such as the use of an uncommon map projection, or alignment such that "up" is not "north")	N/A	N/A	N/A	0
			Counts of manuscripts with acknowledged or reported uncertainty			
			26	28	9	3

Note. NA = not applicable. Each manuscript may have multiple figures, each of which may have been found to contain multiple forms of uncertainty. Total counts refer to manuscript-level counts (e.g., if a manuscript reports both spatial and statistical uncertainty in text, it will count for 1 in those categories, but then only count 1 toward "present in text only" total). See supplementary materials for full data as extracted.

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3.2. Target audience and policy relevance

Approximately 52 out of 66 studies (approximately 79%) clearly mentioned policymakers, public health authorities, or health practitioners as key audiences for their research findings.

These studies contributed to policy relevance by informing targeted interventions and resource allocation and enhancing disease surveillance accuracy through improved spatial prediction models to support control and elimination programs. They also provided actionable insights that support national or regional disease elimination goals. They offered evidence to update or develop new public health policies and control strategies and enabled rapid identification and response to outbreaks through refined spatial and temporal risk mapping. The diseases addressed where policy relevance was emphasized include malaria, dengue, chikungunya, scrub typhus, hand-foot-mouth disease, visceral leishmaniasis, influenza, and soil-transmitted helminths.

3.3. Data source

Among the 66 studies included for full-text screening, the majority (53%) relied exclusively on secondary data sources, including national surveillance databases, electronic health records, government reports, and remote sensing datasets. Approximately 23% of studies collected primary data directly through methods such as field surveys, sentinel surveillance, clinical interviews, and laboratory-confirmed case detection. The remaining 24% of studies employed a mixed approach, combining both primary data collection and secondary data sources.

Uncertainty in the source of data was classified into low, moderate, and high categories. Thirty-five studies using only secondary data generally exhibited low uncertainty, reflecting standardised collection protocols and established databases. Conversely, fifteen studies based on primary data collection often had higher uncertainty due to variability in data quality, potential reporting biases, poor validity and reliability of research tools and methodological differences. Sixteen mixed data studies typically demonstrated moderate uncertainty.

Common challenges across the included studies included underreporting of cases, missing or incomplete data, inconsistent surveillance methods, and, in some instances, reliance on older datasets where data collection methods were less accurate and reliable. These factors contributed to inconsistencies and potential biases in the data, affecting the quality of geospatial analyses.

3.4. Temporal considerations

The timing of sampling in the included studies showed considerable variability, which impacted the ability to capture seasonal and temporal disease dynamics accurately. Twenty studies had limited sampling periods focusing on peak seasons or specific months, which may bias results by missing seasonal variation. For instance, Hu et al. [22] collected data only for May 2008, justified by the peak case count that month, but this approach risks missing important seasonal fluctuations in hand, foot, and mouth disease (HFMD) incidence. Similarly, in scrub typhus studies, sampling was often skewed towards peak transmission months such as October and November, possibly underestimating cases during off-peaks (e.g., as in Xin et al. [23] and Wu et al. [24]).

Conversely, about 15 studies employed long-term continuous or monthly data collection over several years, which better captured seasonal patterns and temporal trends. For example, Ding et al. [25] analysed 10 years of continuous monthly data for HFMD, allowing a robust understanding of seasonal effects. However, Dewan et al. [26] on visceral leishmaniasis lacked explicit discussion on seasonal variability despite known climatic influences on transmission, indicating a temporal sampling limitation.

Eight studies noted delays between illness onset and diagnosis, especially in rural or high-risk areas, which could blur temporal clustering and affect data reliability. Qian et al. [27] highlighted that rural-urban differences in reporting timeliness could lead to variations in observed transmission patterns.

Allen et al. [28] sampled just before a major chikungunya epidemic in Dhaka, which may not represent risk dynamics during the epidemic itself. This provided valuable baseline data but potentially limited the generalizability of findings to other periods.

Regarding spatial scale, studies ranged from national or provincial-level analyses to fine-scale district, community, or household-level studies. Approximately 50% of studies utilized aggregated data at national or regional scales, providing broad geographic coverage but often with less spatial precision. Around 27% of studies conducted analyses at district or sub-district levels, allowing identification of localised transmission patterns. A smaller subset of about 23% of studies used geo-referenced household or village-level data to achieve high spatial resolution, although such fine-scale data posed challenges regarding data completeness and consistency.

Overall, the variability in sampling timing and spatial scale across studies represents a key challenge in accurately characterising disease dynamics and their environmental drivers.

3.5. Visual depiction of uncertainty

In addition to data source and temporal considerations, 3 broad categories of recurring uncertainty in maps were identified from qualitative analysis of the included 66 studies (table 3).

The majority of the included studies did not explicitly represent uncertainty visually on their maps. Instead, uncertainties were often only mentioned descriptively in the text or in figure captions.

Seventeen studies acknowledged uncertainty qualitatively in the text but did not provide quantitative measures such as confidence intervals or error margins (e.g., Reid et al. [30]). This lack of precise quantification limits the ability to assess the reliability of the spatial estimates.

Eleven studies reported statistical measures of uncertainty, such as standard errors or p-values in the textual descriptions but failed to incorporate these metrics directly onto the spatial visualisations [22,31]. Without visible representations of these uncertainties on the maps themselves, policymakers may misinterpret spatial patterns as more precise or reliable than they truly are, potentially overlooking areas where spatial heterogeneity is uncertain or poorly supported by the data.

Nine studies included uncertainty details in figure captions without visualising them within the figures or maps themselves (e.g., as in Hasan et al. [32], Owada et al. [33], Qian et al. [27]). This disconnects between textual information and spatial representation may result in decision-makers neglecting uncertainty when interpreting the maps for policy formulation.

Only three studies effectively integrated explanatory visual representations of uncertainty into their maps, using techniques such as Bayesian Decision Network (BDN) modelling, spatio-temporal modelling, shading gradients or spatial fuzziness to denote areas of higher uncertainty [34–36]. Such clear depictions are valuable for policymakers, enabling them to identify regions with less reliable data and thus adopt more cautious or targeted interventions.

Seven studies, including Zellweger et al. [37], Edenborough et al. [38] and Akter et al. [39], used aggregated spatial units prone to the Modifiable Area Unit Problem, potentially masking fine-scale variability. No maps displayed clear signs of uncertainty arising from poor visual clarity that could not be better attributed to spatial or statistical uncertainty. Temporal uncertainties were also found in the studies, but these were rarely visualised within spatiotemporal maps (e.g., an omission seen in Fahmi et al. [40], and Trewin et al. [41]). For instance, Trewin et al. [41] used data from 1887-2016. This study reported significant temporal gaps in surveillance, especially during the mid-20th century, and changes in data collection methods from sporadic, paper-based reports to systematic digital surveillance. Additionally, aggregating data across such a broad time period without accounting for methodological differences. The lack of explicit temporal uncertainty depiction in the figure could mislead policymakers regarding the stability and robustness of observed trends over time.

3.6. Concordance between observed and reported uncertainties

The previous section focused on the visual representation of uncertainties in maps. It is equally important to consider how uncertainties are communicated in the accompanying text. Over a third of studies ($n = 26$, 39%) didn't report any uncertainties in their research papers (e.g., as in Murphy. et al. [42], Tewara et al. [43], Viennet et al. [44]). In many of these cases, key details such as confidence intervals, error margins, or sampling limitations were missing from both the text and the maps. This lack of transparent reporting means that although uncertainties existed, they were not clearly communicated to readers. Such omissions can mislead policymakers into assuming that the data is more precise and reliable than it truly is, which could lead to overconfident or potentially flawed decisions.

Thirty-seven studies reported some uncertainties in text (and/or figure captions), but these were incomplete (e.g., seen in Hasan et al. [32]). Not all relevant uncertainties were mentioned by the authors, as they acknowledged sampling limitations but failed to report spatial or temporal uncertainties.

Taken together, these findings indicate that most spatial epidemiological studies do not employ visual methods to convey uncertainty. This insufficiency may foster misplaced confidence in map-based data, increasing the risk of misguided

public health decisions. Therefore, transparent and explicit visualization of uncertainty is critical to support evidence-informed, prudent policymaking.

4. Discussion

This systematic scoping review assessed the peer-reviewed literature regarding climate-related infectious disease in the Asia-Pacific region. The aim was to identify whether current practices in map visualisations support informed, cautious, and evidence-based public health responses. Despite using reliable government and standardised data sources, uncertainty was largely omitted from visualisations. Only three out of the sixty-six studies visualised uncertainty in their maps. Forty-three studies either did not report any uncertainty or only mentioned uncertainty in text without connection to visualisations. Eleven studies reported uncertainty in figure captions, without integrating them visually. Communicated uncertainties can significantly undermine the reliability and interpretability of maps to inform policy and decision-making. This is not a recent concern. A decade ago, the lack of uncertainty visualisation within maps was noted by Kinkeldey et al. [45], and the negative consequences of absent or poorly conceived uncertainty within maps were discussed by Scholz and Lu [46]. Given the ongoing use of maps as key decision-support tools for guiding policy making, our findings indicate a longstanding need to improve current practice of uncertainty visualisation in maps relating to climate-sensitive infectious diseases.

The current results build on previous reviews focused on spatial methods and applications. The data, modelling, and variety of maps found in the included manuscripts align with what was found in Nohr et al.'s [47] extensive evaluation of geospatial methods in health research, and in Auchincloss et al.'s [48] exploration of the utility of geospatial information systems and maps in public health. Despite the availability of advanced visualisation methods such as interactive maps, fuzzy boundary representations, and confidence interval overlays, our systematic scoping review demonstrates that these tools remain significantly underutilised. This underutilisation can be attributed to several interrelated factors. First, there is institutional inertia and a strong reliance on conventional and widely adopted software tools, such as Power BI and Excel, which often lack the flexibility to visually encode uncertainty [49]. These platforms are designed for clarity and usability but do not readily support complex cartographic techniques such as transparency layers or probabilistic shading. Second, many researchers and public health practitioners may have limited training in uncertainty visualisation or may prioritise spatial precision over communicating the limitations of their data [11,21]. Third, policymakers often prefer simplified visuals that appear definitive and actionable. As noted by MacEachren et al. [50] and Kinkeldey et al. [45], uncertainty in maps can be perceived as confusing or as undermining trust in the findings, leading to a preference for clean, unambiguous outputs. This is compounded in circumstances where visualisations depict possible future states of the world, and their relative probabilities of occurring [51]. This desire for simplicity may unintentionally discourage the inclusion of visual uncertainty cues that are essential for fully informed decision-making. Maps support more rapid decision making than other formats of information, such as tables, in part due to their ability to simplify complex information using visual heuristics [52]. The introduction of uncertainty may inherently add complexity and undermine the straightforward heuristics that allow them to be used efficiently in decision support. These barriers to improved visualisation of uncertainty in geographic and geospatial data need to be considered alongside the need for any change in practice to remain intelligible to the desired audience.

4.1. Toward better uncertainty communication

Geospatial scientists, decision scientists, epidemiologists, policy experts, and behavioural scientists should work together to ensure that uncertainty is not just technically calculated but also meaningfully interpreted and applied [45,53]. A fruitful first step would be to increase efforts to raise policymaker awareness of the value of understanding uncertainty when prioritising the information sources and focusing interventions and further data collection. There are some examples of this. Degeling et al. [54] outline of how such training may be used to enhance policymaker decision making in disease surveillance, and Wiggins et al. [55] present a decision uncertainty toolkit to support decision making during public health

emergencies. Once the value of uncertainty is understood, the value of visualising uncertainty may be more salient to policymakers. This could be supported by an expectation from journals, reviewers, and agencies commissioning reports that uncertainty will be routinely integrated into policy-focussed reporting, ideally within visuals as well as text. Involving public health stakeholders from the outset ensures maps meet both the informational needs and cultural expectations of decision-makers [56,57].

There is substantial variability in individual's decision performance when using maps [58]. Usability testing, where policymakers interact with maps and provide feedback, can reveal how uncertainty representations such as shaded risk regions, error bars, confidence interval overlays, fuzzy boundaries, probabilistic shading, and ensemble-map variability layers are interpreted. This may require repeated specific co-design, exemplified by Scholz & Lu [46], who emphasise the importance of context-specific maps that explicitly explain uncertainties, modelling assumptions, and data limitations to decision-makers. As a potential way forward, Hadjimichael [59] proposes an exploratory modelling and decision-making framework that integrates scenario discovery with stakeholder engagement. This approach allows uncertainties to be visualised in a decision-relevant manner, enabling policymakers to navigate trade-offs without being overwhelmed by complexity. Such structured participation can support more transparent and actionable geospatial communication in public health, and help to mitigate the endemic tendency toward incredible certitude [15] in policy. An excellent example of this approach, and consequently useful recommendations for best practice is Ślusarski and Jurkiewicz's (2019) [60] exploration of depicting positional, attribute, and temporal uncertainty in Poland's Database of Topographic Objects.

4.2. Strengths and limitations

This systematic scoping review has several strengths, as well as limitations. A major strength of this work lies in its inclusive approach which allowed us to capture a broad spectrum of study types, modelling approaches, and uncertainty communication practices. Building on prior reviews, which tend to focus on text or visuals in isolation, we assessed the representation of uncertainty in text, figure captions, and maps. This level of granularity enabled us to identify subtle omissions such as confidence intervals reported in text but not present in figure captions or displayed on maps. Another notable strength is the wide temporal coverage: we examined all eligible literature from the earliest available records to 2025. However, we included only English-language publications, which may underrepresent research from non-English-speaking countries. Our inclusion criteria was geographically restricted to the Asia-Pacific region, and findings may not be generalizable to other regions with different health system structures or data availability. Relatedly, this manuscript has focussed on uncertainty from a specifically Western epidemiological lens, and using a largely qualitative approach. Objectivity in data and visualisation practice is to some degree socially constructed [61,62]. As Crampton [63] notes, maps embed social values and uncertainties as well as data uncertainties. This may have been compounded by our approach of relying on a single extractor for most data with a randomised double screening of fifteen percent of publications, which is common practice for systematic scoping reviews, but may have introduced some subjectivity. Future research should consider this social context alongside inclusion of policymakers and experts in efforts to improve depiction of uncertainty in practice. A source of insight for this may be citizen science data practises, such as the integration of insights from large "macro" datasets with citizen input on the "micro" level in order to increase data accuracy and usefulness for policy as described in Soltani et al. [64]. Finally, although we limited the scope to climate-sensitive infectious diseases, the explicit focus on uncertainty in visualisations expands the generalisability of results to other health domains, such as non-communicable diseases or injury surveillance.

5. Conclusions

Effective geospatial maps depend not only on the quality of data and modelling approaches but also on transparent representation of uncertainty. This systematic scoping review highlights an ongoing and critical gap in how uncertainty is conveyed in spatial environmental epidemiology studies, especially those addressing climate-sensitive infectious diseases in the

Asia-Pacific region. While geospatial data collection methods, datasets, and technologies have advanced rapidly and are increasingly used to inform public health policy, there is a persistent lack of explicit communication of uncertainty in data and modelling. This lack of transparency risks misinterpretation, misplaced confidence, and suboptimal policy responses.

Our findings underscore an urgent need to report uncertainties and standardise uncertainty communication in spatially based health research, through visual tools, textual explanation, and co-design with policy end-users. Doing so will not only enhance the trustworthiness of spatial outputs but also ensure that maps serve as responsible tools for evidence-informed decision-making. Moving forward, future research must prioritise clarity, transparency, and collaboration to fully realise the potential of maps in advancing public health.

Supporting information

S1 File. Data.
(DOCX)

S2 File. PRISMA checklist.
(DOCX)

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